PORTFOLIO CASE STUDY · RETAIL ANALYTICS

Customer shopping behavior analysis

An end-to-end evaluation of 3,900 retail transactions across 18 features—uncovering customer segments, product performance, discount dynamics, and subscription economics.

3,900 orders | *18 attributes* | *Python + SQL*

Dataset scope: 3,900 purchases across demographics, product categories, and fulfillment.



DATA PREPARATION & PIPELINE

From raw transactions to a structured SQL database

Initial exploration in Python verified schema health across 3,900 records, resolved null values, standardized naming, and loaded the curated table into PostgreSQL.

01 CLEANING

Median imputation

Imputed 37 missing `review_rating` values using the median rating for each specific product category.

02 STANDARDS

Snake case schema

Renamed all 18 headers to clean snake_case for seamless querying and relational documentation in PostgreSQL.

03 DEDUPLICATION

Redundancy drop

Audited `discount_applied` vs. `promo_code_used`; after verifying matching states, dropped the redundant promo column.

04 ENGINEERING

Engineered cohorts

Constructed binned `age_group` cohorts; derived `purchase_frequency_days` before loading data into PostgreSQL.

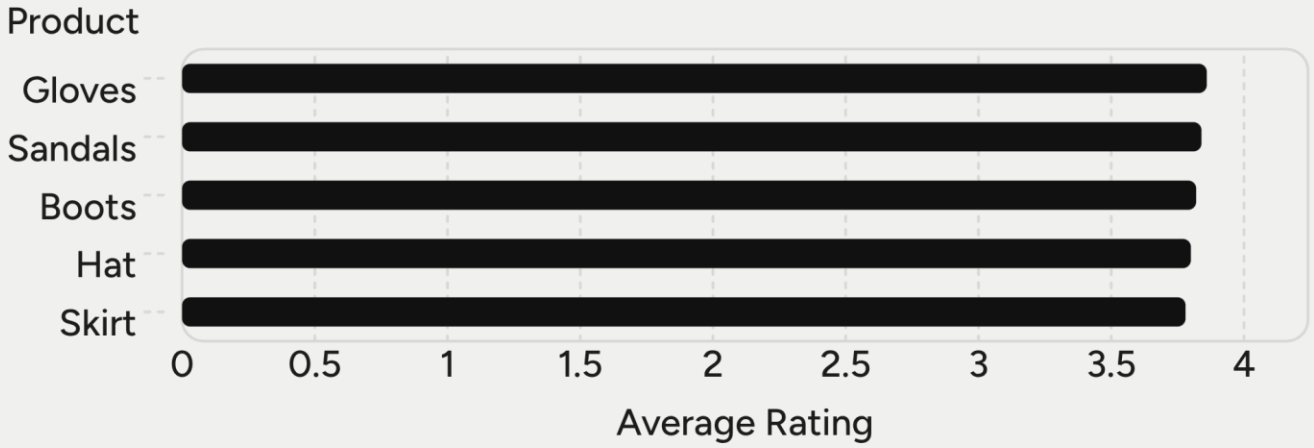
Method: pandas exploration via df.info() and .describe(), transforming data for transactional SQL queries.

SPENDING & SATISFACTION BENCHMARKS

Discounts attract high-volume buyers, led by accessories

Male customers generated $157,890 across 2,652 transactions compared to $75,191 from female buyers. Meanwhile, 839 customers redeemed discounts while still spending above the $59.76 storewide average.

TOP 5 PRODUCTS BY REVIEW RATING



REVENUE SPLIT

$157.9K

Male customers contributed 67.7% of total revenue ($157,890 vs. $75,191 from female buyers).

DISCOUNT ENGAGEMENT

839

High-value discount users maintained baskets above the $59.76 baseline despite price reductions.

✱ SQL queries evaluated gender revenue totals, discount basket thresholds, and top average product ratings.

FULFILLMENT & MEMBERSHIP ECONOMICS

Order values hold steady regardless of shipping or subscription

Average order sizes are virtually invariant across delivery speed and membership tier; subscription volume is the core revenue driver.

Shipping type	Average spend
Express Shipping · High intent	**$60.48 · Premium speed**
Standard Shipping · Baseline	**$58.46 · Default fulfillment**

Subscription status	Customers · Average spend · Total revenue
Subscribers (Yes)	1,053 · **$59.49** · $62,645.00
Non-Subscribers (No)	2,847 · **$59.87** · $170,436.00

Takeaway: Subscribers spend the same per transaction ($59.49 vs $59.87), meaning growth requires increasing membership volume.

Statistical finding: Shipping choices show minimal spend deviation (+$2.02 for Express), while 73% of revenue currently sits with non-subscribers.

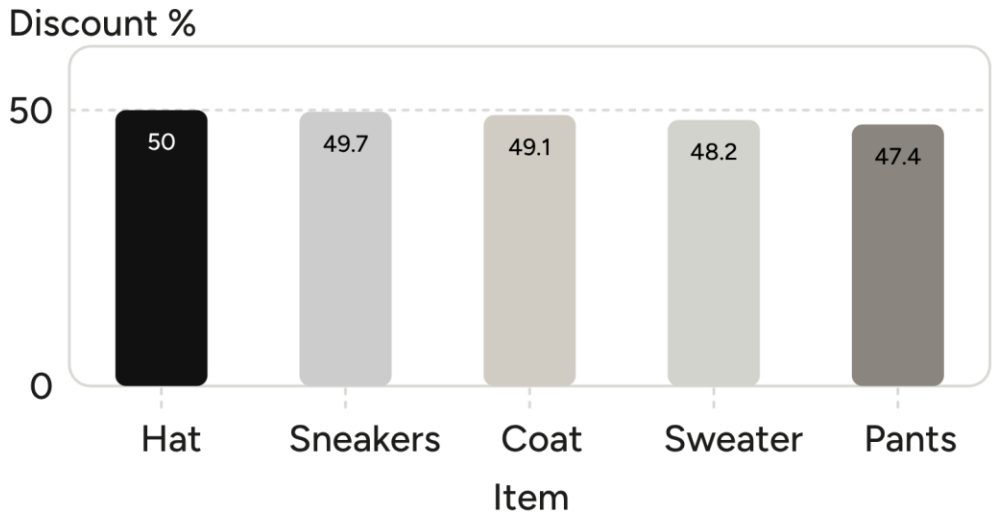
MERCHANDISING & COHORT STRUCTURE

Apparel relies heavily on discounts, while 80% of buyers are loyal

DISCOUNT-DEPENDENT PRODUCTS

Top 5 items by discount rate

Half of all purchases in these high-volume core lines rely on price reductions.



CUSTOMER SEGMENTATION

Purchase history breakdown

Classified from historical purchases: New (83), Returning (701), and Loyal (3,116).

LOYAL — 3,116 customers (79.9%); established long-term buying history.

RETURNING — 701 customers (18.0%); repeat shoppers ripe for targeted loyalty acceleration programs.

NEW — 83 customers (2.1%); first-time buyers requiring dedicated onboarding to secure second orders.

✱ Insight: Margin risks concentrate in Hats and Footwear; retention strategy should focus on migrating 701 Returning buyers into Loyal tiers.

CATEGORY LEADERS & REPEAT BUYERS

Clothing and Accessories drive volume across balanced cohorts

TOP 3 PRODUCTS PER CATEGORY

Category	Top Ranked Items	Orders
Clothing	Blouse (171), Pants (171), Shirt (169)	**511 total**
Accessories	Jewelry (171), Sunglasses (161), Belt (161)	**493 total**
Outerwear	Jacket (163), Coat (161)	**324 total**
Footwear	Sandals (160), Shoes (150), Sneakers (145)	**455 total**

REPEAT BUYERS (>5 PURCHASES)

958 subscribers

Among frequent repeat buyers, **958 hold active subscriptions** while 2,518 remain non-subscribers—representing a major conversion opportunity.

REVENUE BY AGE GROUP

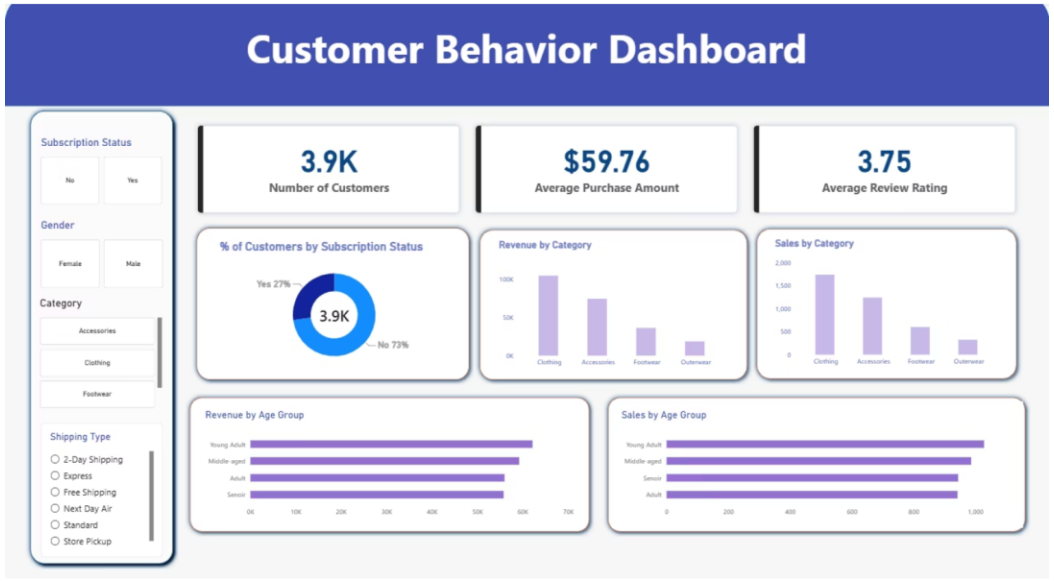
Young Adult · **$62,143** · Middle-aged · **$59,197** · Adult · **$55,978** · Senior · **$55,763**

Revenue distribution is steady across demographics, peaking with Young Adults at $62.1K; 2,518 repeat non-subscribers form the primary pipeline.

POWER BI DASHBOARD DELIVERABLE

An interactive BI environment for executive monitoring

The complete transactional dataset was deployed to an interactive Power BI dashboard, enabling cross-filtering across demographic segments, fulfillment options, and category trends.



DASHBOARD CORE METRICS

TOTAL CUSTOMERS

3.9K

3,900 total logged transactions

AVG PURCHASE AMOUNT

$59.76

Stable baseline basket size

AVG REVIEW RATING

3.75 / 5.0

High overall product satisfaction

Deliverable: Interactive Power BI report providing category-level and demographic slicing for commercial leadership.

STRATEGIC ACTION PLAN

Five data-backed moves for revenue & retention

SQL query results and dashboard findings guide business moves for commercial growth and margin protection.

Boost subscription perks
Promote exclusive benefits and free shipping to 2,518 repeat non-subscribers who already trust the catalog.

Nurture returning buyers
Automated second-purchase incentives to convert 701 "Returning" customers into the high-retention "Loyal" cohort.

Calibrate discount policy
Re-evaluate markdown thresholds on Hats (50%) and Footwear (~49%) to defend gross margins without depressing volume.

Hero top-rated products
Feature top-reviewed lines (Gloves, Sandals, Boots) and high-volume staples (Blouses, Jewelry) in hero acquisition campaigns.

Target high-yield cohorts
Direct premium acquisition campaigns toward Young Adults ($62.1K revenue) and promote Express shipping upgrades ($60.48 average spend).

Executive takeaway: Prioritize subscription conversions among existing repeat buyers and curb heavy markdowns on core apparel lines.